

Sep 9, 2021

Introduction to Computer and Network Security

Part 1

Question 1 [35 points]

In the context of TLS, (a) draw the message sequence chart of the handshake protocol and briefly describe each step, (b) explain how TLS provides for authentication, confidentiality, and integrity.

Question 2 [15 points]

Define the notions of vulnerability and threat and give (at least) an example for each one.

Part 2

Question 3 [35 points]

In the context of data protection and privacy, (a) give two complementary definitions of privacy and briefly comment on them, (b) explain the linkage attack on anonymized dataset with an example, and (c) explain the trade-off between anonymity and utility in the context of realizing data sets.

Question 4 [15 points]

Explain the framework of Attribute Based Access Control (ABAC), in particular how an access request is processed, and the reasons for which it has been introduced with respect to previous models (such as RBAC).

Please, remember...

On each sheet, make sure to write: Name Surname Student ID number Question number	The file containing the scan of your answers should be named as follows: surname-name-studentIDnumber.pdf
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